Here is how a real database changes at every step of normalization, using visual tables.

---

1. First Normal Form (1NF)

The Problem: Cells contain multiple values (a list of hobbies separated by commas).

- **Unnormalized Table:**

StudentID	StudentName	Hobbies
101	Alex	Swimming, Reading
102	Ryan	Football

- **Normalized (1NF) Table:** Each cell now has exactly **one** value. [1]

StudentID	StudentName	Hobby
101	Alex	Swimming
101	Alex	Reading
102	Ryan	Football

---

2. Second Normal Form (2NF)

The Problem: The primary key is a composite key **[OrderID + ProductID]**. The BrandName column only depends on ProductID (part of the key), not the OrderID. [2]

- **1NF Table (with 2NF violation):** [3]

OrderID (Key)	ProductID (Key)	Quantity	BrandName
5001	P99	2	Apple
5001	P88	1	Samsung
5002	P99	5	Apple

- **Normalized (2NF) Tables:** We split them so product details live in their own home.**Orders Table:**

OrderID	ProductID	Quantity
5001	P99	2
5001	P88	1
5002	P99	5

- **Products Table:**

ProductID	BrandName
P99	Apple
P88	Samsung

---

3. Third Normal Form (3NF)

The Problem: ManagerName depends on Department, and Department depends on EmployeeID. This is a chain reaction (transitive dependency). If a department has no employees, we lose the manager's name entirely. [4]

- **2NF Table (with 3NF violation):**

EmployeeID (Key)	EmployeeName	Department	ManagerName
E01	Alice	IT	John
E02	Bob	HR	Sarah
E03	Charlie	IT	John

- **Normalized (3NF) Tables:** We break the chain.**Employees Table:**

EmployeeID	EmployeeName	Department
E01	Alice	IT
E02	Bob	HR
E03	Charlie	IT

- **Departments Table:**

Department	ManagerName
IT	John
HR	Sarah

---

4. Boyce-Codd Normal Form (BCNF)

The Problem: The key is **[StudentID + Subject]**. Each professor teaches only one subject, but a subject can have many professors. Professor determines Subject, but Professor is not a candidate key. [5, 6]

- **3NF Table (with BCNF violation):**

StudentID (Key)	Subject (Key)	Professor
1001	Math	Prof. White
1001	Physics	Prof. Green
1002	Math	Prof. Black
1003	Math	Prof. White

- **Normalized (BCNF) Tables:Student_Professor Table:**

StudentID	Professor
1001	Prof. White
1001	Prof. Green
1002	Prof. Black
1003	Prof. White

- **Professor_Subject Table:**

Professor	Subject
Prof. White	Math

Professor	Subject
Prof. Green	Physics
Prof. Black	Math

To help you lock down this knowledge, tell me:

- Would you like to try **normalising a sample table** yourself right now?
- Or should we focus on how to **join these tables back together** using SQL queries?

[1] https://www.coursehero.com
[2] https://www.red-gate.com
[3] https://www.mygreatlearning.com
[4] https://levelup.gitconnected.com
[5] https://www.red-gate.com
[6] https://study.com